NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

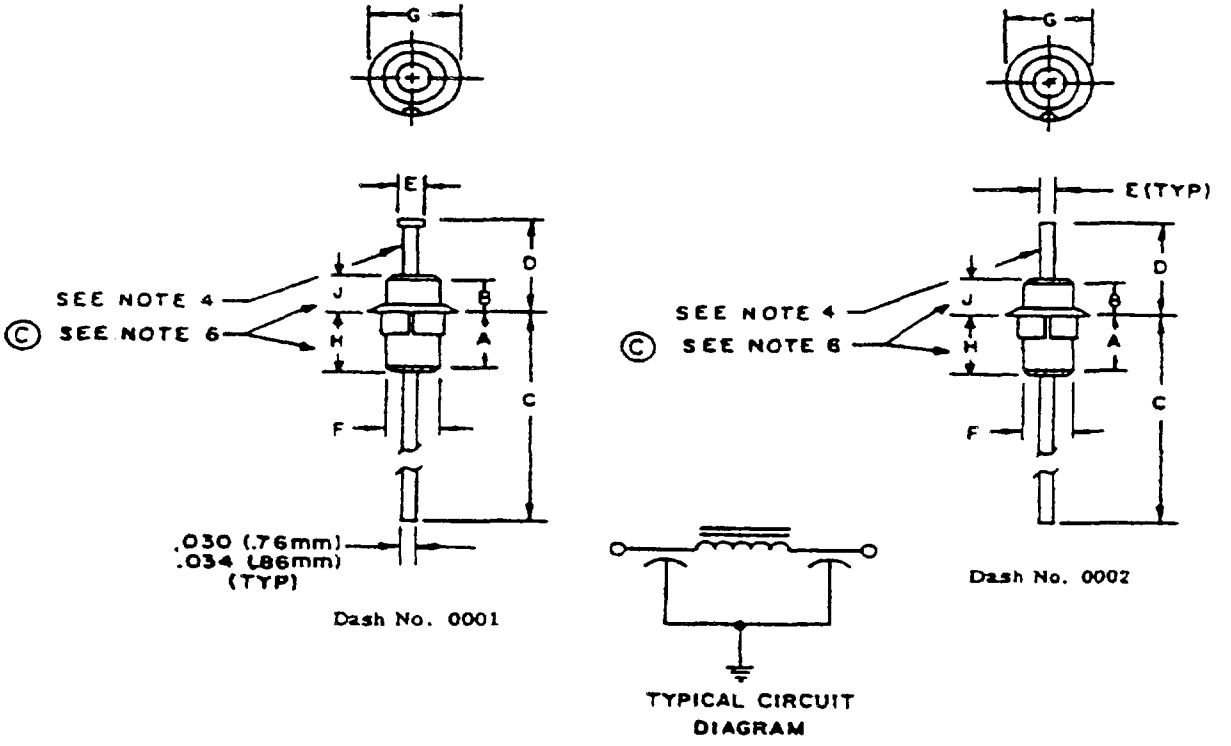
MIL-PRF-15733/33C
27 June 1988
SUPERSEDING
MIL-F-15733/33B
2 November 1977

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, STYLE FL38

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram for information only.
4. Leads shall be solid, tinned, AWG 20 (.032).
5. © Suggested mounting hole diameter, .160 (4.06mm).
6. Dimensions H and J are from mounting surface to clean lead.

FIGURE 1. Dimensions and configuration.

© denotes changes

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REQUIREMENTS:

Configuration and dimensions: See figure 1 and table I.

Weight: 1.1 grams, maximum.

Case: Not applicable (metallic mounting eyelet).

Terminals: Solderable.

Operating temperature range: -55°C to +125°C; temperature characteristic D.

Rated voltage: 125 V dc or 90 V rms, to 400 Hz over the operating temperature range of -55°C to +125°C.

Rated current: 10 amperes, dc or ac (rms).

© Insertion loss characteristic: See table II.

Seal: Not applicable.

Solderability: Method 208, MIL-STD-202. Number of terminations to be tested shall be 2.

Capacitance to ground: Method 305, MIL-STD-202. Test frequency: 1,000 Hz ±100 Hz. Measured capacitance shall be at least 1,750 pF.

Temperature rise: 25°C maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.

Test voltage: 750 V dc applied for 1 to 5 seconds.

Maximum charging current: 50 mA.

Barometric pressure (reduced):

Method 105, test condition D, MIL-STD-202.

Dielectric withstanding voltage: In accordance with initial requirements, except that the test voltage shall be 125 percent of the rated dc voltage.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Insulation resistance measured at 25°C between either terminal and the case shall be at least 10 G ohms.

Voltage drop: Not applicable.

Insertion loss: In accordance with MIL-F-15733.

Overload: In accordance with MIL-F-15733.

Measurements at +25°C after test:

Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A, MIL-STD-202.

Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202

Thermal shock: Not applicable.

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Immersion: Not applicable.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Electrical load: 100 percent of rated voltage between terminals and case.

Measurement after test: Insertion loss (check test) shall meet initial requirements.

Vibration, high frequency: Method 204, test condition D, MIL-STD-202.

Electrical load: Rated voltage, and rated current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the temperature of step 3 shall be +125°C, +3°C -0°C. No measurement shall be made before and after cycling.

Filters shall then be tested in accordance with method 106, MIL-STD-202.

Initial measurement: Not applicable.

Loading voltage: Not applicable.

Polarization voltage: Not applicable.

Measurements after 24-hour drying period, at a temperature not to exceed +85°C, and a relative humidity of 50 percent.

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 1 G ohm.

Insertion loss (check test) shall meet initial requirements.

Life: Method 108, test condition D, MIL-STD-202. Test temperature +125°C.

Test voltage: 1.2 times rated rms voltage.

Test current: 10 amperes.

Measurements after test:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 1 G ohm.

Insertion loss (check test) shall meet the initial requirements.

Marking: Filters shall not be marked. Full marking in accordance with MIL-F-15733, shall be marked on the unit package.

Part number: M15733/33- (dash number from table I).

Supersession data: This specification includes the requirements of contractor (35351) drawing number 150247, dated 28 September 1972.

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© TABLE I. Filter dimensions.

Dash no.	A		B		C		D		E		F	G		H	J
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max	Min	Max	Max	Max
0001	.142 (3.61)	.197 (5.00)	.078 (1.98)	.108 (2.74)	.749 (19.02)	.811 (20.60)	.273 (6.93)	.303 (7.70)	.045 (1.14)	.075 (1.91)	.125 (3.18)	.175 (4.45)	.205 (5.21)	.245 (6.22)	.155 (3.94)
0002	.142 (3.61)	.197 (5.00)	.078 (1.98)	.108 (2.74)	.749 (19.02)	.811 (20.60)	.273 (6.93)	.303 (7.70)	.030 (.76)	.034 (.86)	.125 (3.18)	.175 (4.45)	.205 (5.21)	.245 (6.22)	.155 (3.94)

TABLE II. Insertion loss versus frequency.

Dash number	Minimum insertion loss (dB) in accordance with MIL-STD-220, at +25°C				
	50 MHz	100 MHz	200 MHz	500 MHz	1-10 GHz
0001*	35	50	50	60	60
0002*	35	50	50	60	60

* Preferred for use in the design of equipment.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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